

WASP-52b

R-1319

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_201013 · created 2026-07-31T19:23:41+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459135.754 +/- 0.0058 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.202 +/- 0.061
Transit depth (Rp/Rs)^2	4.07 +/- 2.46 [%]
Orbital Inclination (inc)	83.49 +/- 2.76
Airmass coefficient 1 (a1)	0.25 +/- 0.17
Airmass coefficient 2 (a2)	1.29 +/- 0.72
Scatter in the residuals of the lightcurve fit is	66.55 %
Best Comparison Star	#1 - [351, 187]
Optimal Aperture	0.75
Optimal Annulus	6.0
Transit Duration (day)	0.055 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.202 ± 0.061 vs 0.1646 (deviation 0.5σ)
Duration fit vs published	0.055 d vs 0.0758 d (deviation 0.56σ)
Mid-time O–C	-13.9 min
Depth SNR	2.7
Residual scatter	66.55 %

Light curve

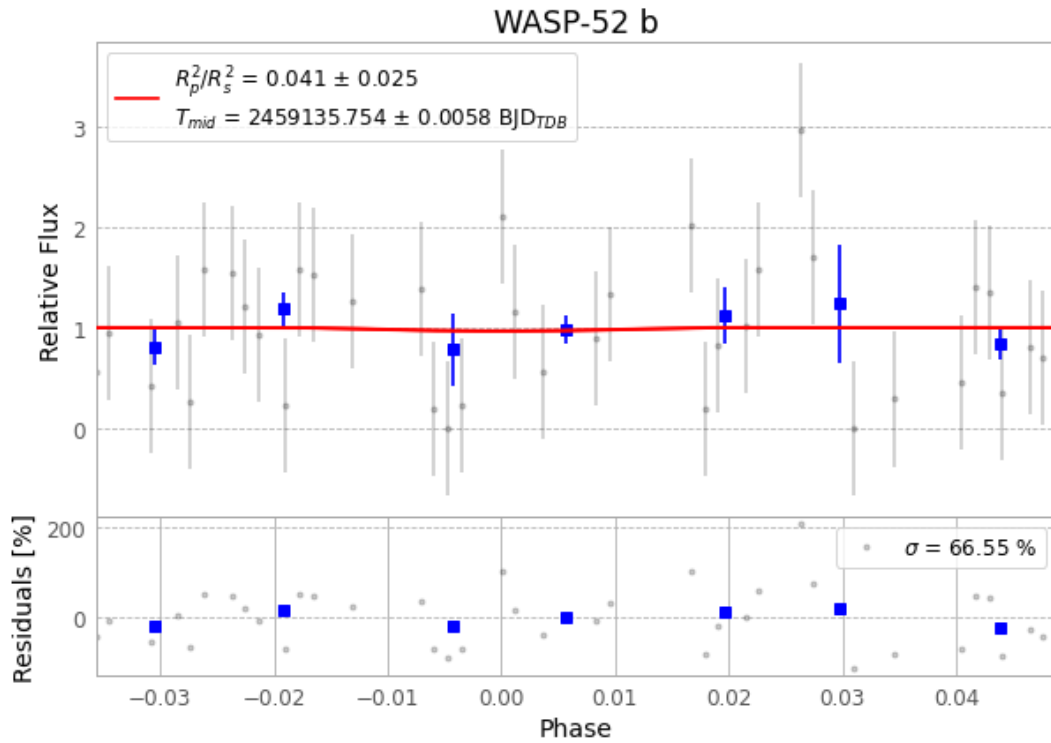


Figure 1 — FinalLightCurve_WASP-52 b_2020-10-12.png (SHA-256 98a45858f0875f64...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [592, 383], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 42c585c3eb8e7cc4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-52201013042412.FITS → WASP-52201013081212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-201013.log (4b7575414bc7...), FinalLightCurve_WASP-52 b_2020-10-12.png (98a45858f087...), FinalLightCurve_WASP-52 b_2020-10-12.csv (e64c4451808c...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 19:23Z.